

ENNA, Italy

WMO: 164500

Lat: 37.568N

Lon: 14.280E

Elev: 3291

StdP: 13.03

Time Zone: 1.00 (EUC)

Period: 02-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	30.2	32.3	7.8	9.2	53.6	15.4	13.5	53.1	36.8	45.2	29.7	40.9	9.2	300	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.5	90.5	65.2	87.6	64.2	84.8	63.2	73.6	80.7	71.6	78.7	69.6	76.8	5.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.8	133.0	76.0	69.7	123.4	75.0	67.6	114.8	73.0	39.7	80.4	37.7	77.5	35.9	76.4	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.8	18.8	15.7	DB	26.5	96.4	2.7	5.0	24.5	100.0	23.0	102.9	21.5	105.8	19.6	109.4	(4)
(5)			WB	24.8	75.6	2.4	5.1	23.0	79.3	21.6	82.3	20.2	85.1	18.5	88.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.2	43.6	42.7	46.3	53.0	60.8	70.1	76.1	76.1	67.8	61.3	52.6	46.7	(6)	
	DBStd	13.28	5.47	5.70	6.32	6.51	6.83	7.62	5.35	5.76	6.29	5.89	5.59	4.89	(7)	
	HDD50	777	208	214	151	42	2	0	0	0	0	1	36	123	(8)	
	HDD65	3558	664	624	579	364	169	33	1	1	37	146	374	566	(9)	
	CDD50	3765	9	10	36	130	336	603	810	808	535	352	114	22	(10)	
	CDD65	1072	0	0	0	3	38	186	346	344	122	32	1	0	(11)	
	CDH74	6834	0	0	0	4	155	1143	2482	2437	550	61	2	0	(12)	
	CDH80	2110	0	0	0	0	25	352	807	799	123	4	0	0	(13)	
(14) Wind	WSAvg	6.1	6.7	7.5	7.3	6.9	5.8	5.2	4.7	5.4	5.7	5.5	6.5	6.0	(14)	
(15) Precipitation	PrecAvg	17.3	2.3	1.9	1.8	1.3	0.9	0.4	0.3	1.2	1.5	2.9	2.2	2.4	(15)	
	PrecMax	29.8	9.4	6.9	7.2	4.1	3.7	2.2	2.4	10.0	4.5	8.4	10.4	6.2	(16)	
	PrecMin	5.4	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.1	(17)	
	PrecStd	6.7	2.2	1.7	1.7	1.0	0.9	0.6	0.5	2.0	1.0	2.1	2.1	1.8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.1	61.2	65.8	72.0	83.9	92.1	94.7	94.9	88.0	79.0	68.4	63.0	(19)	
		MCWB	47.6	48.1	51.5	55.5	57.0	61.8	69.4	68.0	67.5	59.1	52.7	51.6	(20)	
	2%	DB	55.0	56.4	61.0	68.2	78.1	87.6	90.4	90.0	83.3	75.3	64.6	58.7	(21)	
		MCWB	44.4	45.9	50.0	52.9	56.0	60.5	65.3	66.6	64.0	58.6	52.8	51.5	(22)	
	5%	DB	52.1	53.3	57.6	64.7	74.8	84.1	87.2	87.4	79.8	71.8	61.7	54.8	(23)	
		MCWB	44.6	45.4	47.5	51.4	56.0	60.8	64.2	65.3	63.2	58.0	53.6	50.3	(24)	
	10%	DB	50.0	50.2	55.1	62.2	70.5	80.5	84.4	84.5	76.8	68.3	59.1	52.0	(25)	
		MCWB	44.9	44.3	46.8	50.2	54.7	60.4	63.1	64.0	62.9	57.7	53.8	48.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.0	52.8	58.1	59.9	66.2	71.7	76.7	76.4	74.7	68.0	61.6	56.9	(27)	
		MCDB	54.7	56.3	59.8	67.2	71.4	79.9	85.5	82.9	79.8	70.7	62.8	58.7	(28)	
	2%	WB	49.7	50.0	53.5	57.1	62.6	68.9	73.5	73.7	71.5	65.2	59.1	53.9	(29)	
		MCDB	51.1	52.6	56.4	62.9	69.0	77.0	81.4	81.0	76.4	68.6	60.1	55.9	(30)	
	5%	WB	48.2	48.1	50.4	55.1	60.1	66.5	70.9	71.8	68.9	63.4	57.3	51.8	(31)	
		MCDB	49.4	50.2	54.1	60.5	66.5	75.3	77.9	79.9	73.5	66.2	58.6	53.3	(32)	
	10%	WB	46.7	46.3	48.5	53.1	58.1	64.3	68.7	69.9	66.8	61.8	55.6	50.3	(33)	
		MCDB	48.0	48.3	52.1	58.0	65.4	73.3	76.5	77.9	71.2	64.4	56.9	51.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.7	8.7	10.3	11.7	13.8	14.9	15.5	15.7	13.1	10.6	8.2	7.2	(35)	
		MCDBR	10.7	12.8	14.4	15.5	16.8	16.7	17.1	17.2	15.6	14.4	11.1	10.3	(36)	
	5% WB	MCWBR	8.6	9.9	11.7	11.3	11.0	9.6	10.0	10.1	10.5	10.2	8.8	8.0	(37)	
		MCWBR	8.7	10.5	12.5	14.0	14.8	16.0	16.4	16.8	14.4	12.3	9.1	8.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.308	0.328	0.367	0.403	0.404	0.387	0.383	0.391	0.391	0.368	0.339	0.314	(40)	
		taud	2.473	2.397	2.275	2.183	2.222	2.299	2.315	2.292	2.312	2.366	2.419	2.471	(41)	
	Ebn at Noon		279	286	283	278	279	283	283	279	272	267	263	268	(42)	
		Edh at Noon	26	32	39	46	45	42	41	41	37	32	27	24	(43)	
(44) All-Sky Solar Radiation	RadAvg		709	953	1353	1734	2090	2351	2437	2135	1587	1152	797	640	(44)	
	RadStd		60	99	111	128	119	116	83	123	79	84	59	58	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	-0.67	-0.85	N/A	N/A	N/A	N/A

Nomenclature: See separate page